"DEVELOPMENT OF A SYSTEM FOR THE IDENTIFICATION OF LUNG DISEASES BASED ON METABOLOMIC METHODS IN VULNERABLE COMMUNITIES IN SAN LUIS POTOSÍ, MEXICO"

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1 INTRODUCTION

In Mexico, artisanal brick production is a significant economic activity that provides employment for more than 15,000 families, but it operates largely in the informal sector, with precarious working conditions such as a lack of formal contracts, long working hours, low wages, limited access to healthcare and a lack of personal protective equipment or respiratory protection. Furthermore, the use of low-cost fuels for brick firing, combined with technologically inefficient kilns, contributes to the emission of potentially toxic pollutants, such as polycyclic aromatic hydrocarbons (PAHs), particulate matter (PM_{10} and $PM_{2.5}$), heavy metals, dioxins and furans, posing a serious risk to the environment, the public health of workers, their families and neighbouring communities.

2 OBJECTIVE

To assess changes in lung function and respiratory risk amongst brickmakers exposed to environmental pollutants in San Luis Potosí (Mexico), before and after the implementation of a multidimensional intervention, using spirometry tests, metabolomic tools and a questionnaire on respiratory health.

3 METHODOLOGY

STUDY AREA:

San Luis Potosí, Mexico.

MATERIALS AND METHODS:

Lung health questionnaire

Evaluation of pulmonary function:

Spirometer EasyOne

Breath test:

Electronic nose Cyranose 320

Statistical Analysis:

Shapiro-Wilk, ANOVA, GraphPad Prism, SIMCA® Multivariate data analysis

STUDY DESIGN:

Before intervention → Intervention → After intervention

Phase 1. Before intervention

Spirometry test → Lung health questionnaire → Metabolomic test

Phase 2. Intervention

Health education; Workshops; Talks

Coming soon

Phase 3. Post-intervention

Epidemiological monitoring of lung function among brickmakers

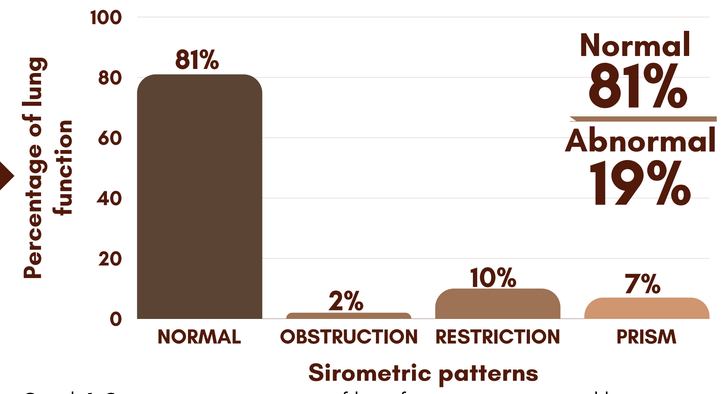
4 RESULTS

N=42

1. Assessment of lung function by spirometry



RESULTS PHASE 1



Graph 1. Spirometric parameters of lung function as measured by spirometry

2. Implementation of lung health questionnaire



Table 1. Results of the lung health questionnaire and the respiratory risk

	NORMAL LUNG FUNCTION	ABNORMAL LUNG FUNCTION
N=42	34 (81%)	8 (19%)
*Type of expositions (out of 5)	4	3
Smoking	3	1
**Index of exposure to firewood or biomass	352 hrs firewood/ year	216 hrs firewood/ year
Respiratory symptoms (over the last 3 months)	85%	90%
Personal and family medical history of Respiratory Diseases	55%	100%

*Exposure to firewood, dust, smoke from brickworks, joinery, and inhalants.
**>200 hrs of firewood/year: Significant increase in the risk of developing COPD.

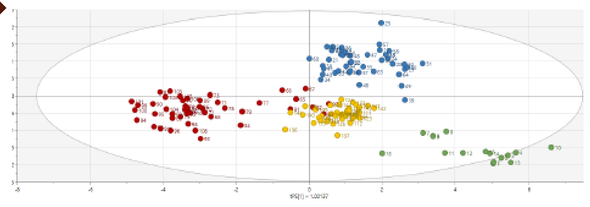
4 RESULTS

RESULTS PHASE 1

3. Metabolomic tests using exhaled breath



Graph 2. Aplication of the metabolomic model of respiratory diseases with supervised analysis: OPLS-DA



72% COPD metabolomic signature

RESULTS PHASE 2

1. Workshops, talks, educational materials and activities on lung health



Graph 3. Workshop on the use of respiratory protection

100% Workers do not have access to healthcare services

100% Workers do not wear respiratory protection

70% Workers are unaware of the lung diseases caused by exposure to environmental pollution

60% Workers are unaware of the importance of respiratory symptoms

5 CONCLUSION

- Over 50 per cent of the population has various risk factors for respiratory diseases
- 19 per cent of brickmakers have impaired lung function
- Metabolomics serves as a tool for the early detection of lung diseases.
- Around 80 per cent of people have a metabolomic profile characteristic of COPD.
- It is important to promote public policies in Mexico that improve working conditions and reduce exposure to pollutants.
- The implementation of multi-dimensional interventions is essential to identify risks and improve workers' living conditions.
- Brickmakers do not use personal protective equipment or respiratory protection and have been working in these conditions for years, meaning they are exposed to thousands of pollutants over long periods of time.
- Training through talks, workshops and educational materials on risk communication is proving to be a powerful tool for raising public awareness of occupational respiratory risks.
- Improvements to working conditions such as decent work, the right to a healthy environment, and the right to health and medical care must be understood as a right and not as a privilege.
- Lung health surveillance in communities exposed to pollutants can be a valuable tool for preventing the risk of acute or chronic respiratory diseases.
- Promoting health through the use of personal protective equipment and respiratory protection, a healthy diet and preventative care can help to protect lung health and improve the quality of life of brickmakers, their families and the surrounding community.
- The use of sustainable fuel materials can help to protect health and the environment

6 REFERENCES

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